

디지털 영상처리(이,공) (EE402-01)

학기	2026년 2학기
담당교수	이병권
이수구분	교선�필수
학점	3.0
강의시간	화10:30-12:00(E7 - L28), 목10:30-12:00(E7 - L28)
강의실	E3 611 (교수실)
소속	기초학부

교과개요

This course provides an in-depth exploration of digital image processing techniques and their applications. Fundamental concepts such as image representation, enhancement, restoration, reconstruction, and segmentation are covered. Students will gain hands-on experience with modern image processing algorithms and techniques, applying theoretical knowledge to real-world problems. The course will also emphasize the mathematical foundations behind digital image operations, preparing students for both academic and industry careers in fields such as computer vision, machine learning, biomedical imaging, and multimedia applications.

교과목표

Understanding key concepts such as image acquisition, image representation, and 2D Fourier transform.
Understanding basic techniques of digital image processing such as spatial filtering, image enhancement, image restoration and reconstruction, morphological image processing, and image segmentation. Exercising some key examples on MATLAB to enhance skills for implementation of those techniques.

주차별 강의계획

- 1주차:Introduction, Review of Linear Systems
- 2주차:Digital Image Fundamentals
- 3주차:Intensity Transformations and Spatial Filtering
- 4주차:Spatial Fourier Transform
- 5주차:Filtering in the Frequency Domain
- 6주차:Image Restoration
- 7주차:Image Reconstruction
- 8주차:Midterm Exam
- 9주차:Morphological Image Processing
- 10주차:Basic Image Segmentation
- 11주차:Advanced Image Segmentation
- 12주차:Feature Extraction
- 13주차:Image Pattern Classification
- 14주차:Image Processing Using Deep Neural Networks
- 15주차:Final Project Presentations
- 16주차:Final Exam

학습윤리

Discussion and collaboration amongst students are encouraged. However, offering and accepting solutions are acts of plagiarism—a serious academic integrity violation—for which all the involved students will be penalized.

교과정책(방침)

Completed homework should be turned in before the beginning of the class on the due date. No late homework will be accepted. Also, students should write their answers as clearly as possible so that those can easily be understood by the grader. No credit points will be given to homework answers which cannot be read or understood. Taking photos is prohibited during class. Instead, lecture recordings will be provided.